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ARTICLE

Wafer-Scale Laser-Writing of Nanoporous Membranes with Monodisperse Pores for Robust Immunoisolation

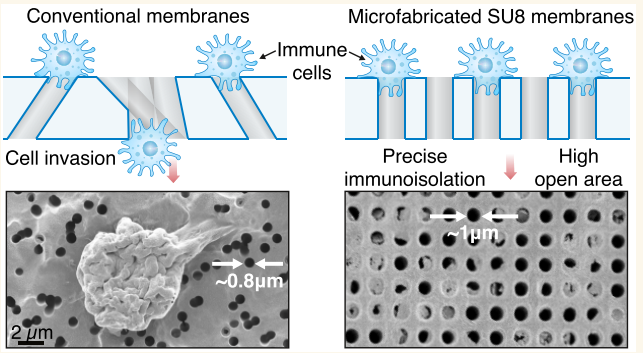
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ABSTRACT: **Membranes that prevent cellular infiltration are essential components of immune-isolation devices. Stringent pore size control is critical to robustly exclude immune cells while preserving the efficient transport of nutrients and therapeutic molecules. Conventional membrane fabrication methods, however, struggle to balance these two properties. For instance, phase separation and fiber spinning generate highly tortuous pores that restrict diffusion, while random pore overlaps due to ion track etching result in oversized defects that compromise immune isolation. Here, we leverage scalable laser-writing tools to achieve spatially controlled open-through and dense pores with uniformly distributed submicrometer sizes for rapid transport and robust immune cell exclusion. Given the optical limits of laser aligners when fabricating submicrometer features, we tuned substrate reflectivity and exposure parameters to overcome these resolution limits. Self-standing and flexible membranes were achieved via thickness-tunable grid layers supporting membrane mechanical integrity. We report pore sizes as small as *ca.* 600 nm (below the resolution limits of the laser exposure systems) and over 20% open area at the exposure times of ∼3 min/cm^2. Compared with conventional methods, this approach minimizes pore tortuosity, narrows the pore distribution, and improves the open pore density. Enhanced pore control is highlighted by macrophage infiltration studies, which indicate that these membranes reliably exclude macrophages at nominal pore sizes greater than those previously achievable. This approach thus provides a scalable pathway toward next-generation immunoisolation membranes for enhanced implantable therapeutic devices.**



KEYWORDS: *Membranes, microfabrication, immunoisolation, direct laser writing, porous*

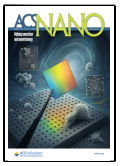
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INTRODUCTION

Nanoporous membranes with precisely controlled pore sizes play essential roles across diverse biomedical applications, including biochemical separation, tissue engineering, and cellular encapsulation.[1−4] For cell encapsulation devices, stringent pore size control is particularly critical, as membranes must robustly exclude immune cells while enabling efficient exchange of nutrients and therapeutic molecules.[5−7] In these immunoisolation applications, the membranes' ability to prevent immune infiltration while maximizing molecular transport directly impacts device longevity and clinical efficacy.[8,9]

Polytetrafluoroethylene (PTFE)-based membranes[10] have been utilized in ViaCyte's PEC-Encap encapsulation devices (acquired by Vertex Pharmaceuticals).[11,12] However, these membranes typically exhibit tortuous pore pathways, limiting molecular diffusivity and oxygen transport.[13−16] Molecular diffusivity limitations apply to traditional polymeric filtration membranes fabricated via nonsolvent-induced phase separation (NIPS) or thermally induced phase separation (TIPS),[17] using polymers such as cellulose acetate,[18] polyvinylidene fluoride (PVDF),[19] and poly(ether sulfone) (PES),[20] and cellulose-based dialysis membranes.[21] Nanoporous ceramic membranes, such as anodized alumina[22] or silicon-based substrates,[23] provide relatively straight pore architectures but remain limited

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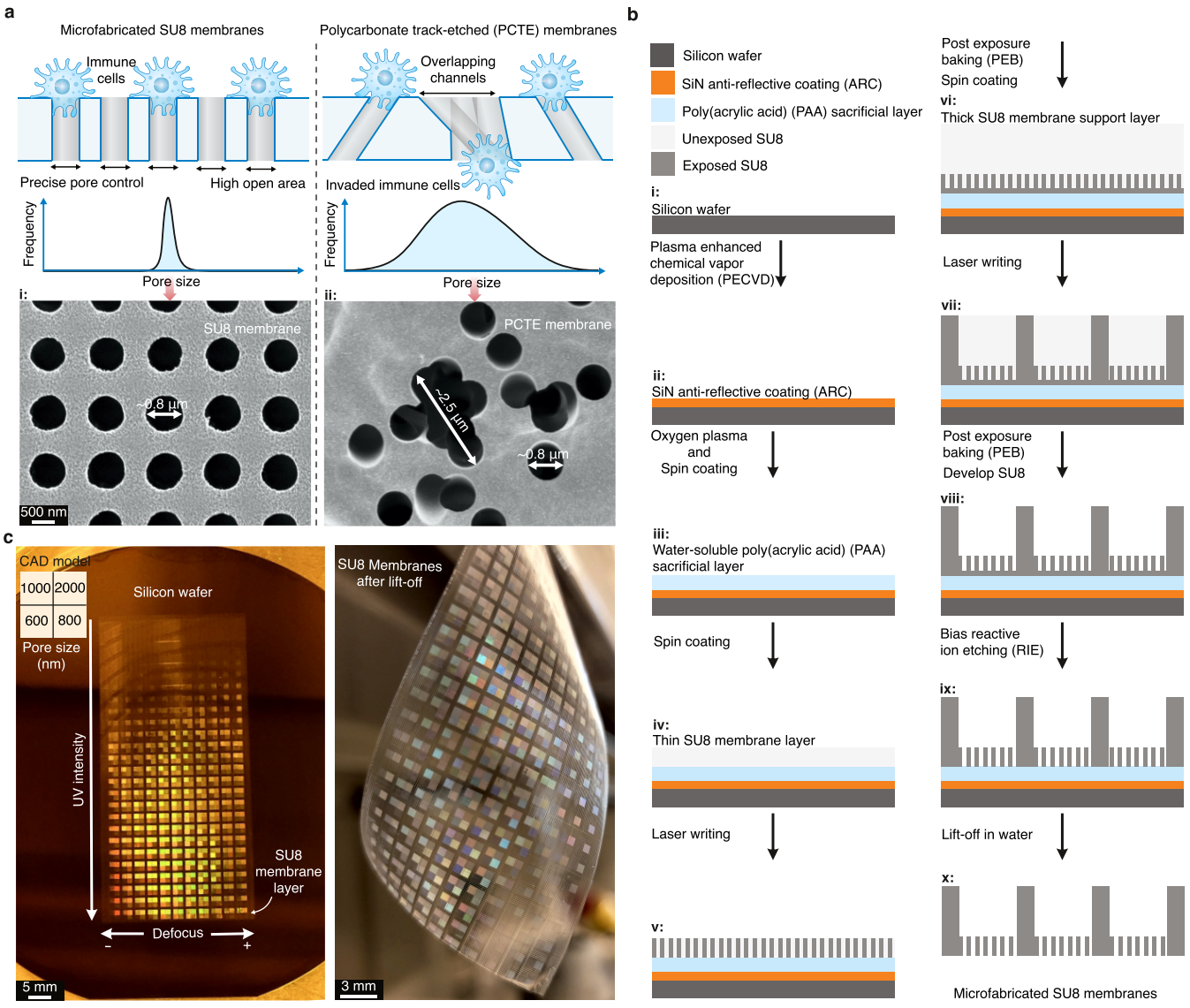


Figure 1. Microfabrication of highly porous SU8 membranes with precise pore control. (a) Pore size control in membranes using microfabrication processes enables efficient isolation of biological components such as immune cells. Scanning electron microscopy (SEM) images show microcontrolled SU8 membranes and commercially available polycarbonate track-etched (PCTE) membranes. (b) Microfabrication of SU8 membranes patterned via a rapid direct laser-writing process. The membranes are mechanically supported by a thick supporting grid frame. (c) Membrane pore control is achieved by computer-aided design (CAD) models and exposure parameters, including UV exposure dose and focal plane adjustments.

by mechanical brittleness and restrictive pore sizes, typically in the tens of nanometers, that significantly reduce exchange kinetics.

Ion track-etching technology provides a compelling alternative, creating relatively straight, through-pores by bombarding polymer membranes with accelerated heavy ions.[24] A recent study using polycarbonate track-etched (PCTE) immunoisolation membranes suggested that the critical pore size for effective immune protection is around 800 nm.[25] However, stochastic ion bombardment in track etching frequently leads to overlapping pores, creating defects exceeding critical size thresholds, which compromise reliable immunoisolation. As such, *in vivo* studies using PCTE cell encapsulations have primarily been carried out with more restrictive 400 nm pores.[25]

Collectively, these challenges highlight the critical need for scalable polymeric membranes featuring straight, uniformly controlled pore structures, improved diffusivity, and precise immunoisolation capabilities to enable more reliable and effective implantable therapeutic systems.[26] Photolithographic microfabrication techniques represent a promising avenue for precise pore size and spatial control.[27] Yet, conventional approaches capable of fabricating pores at submicron dimensions, including electron-beam lithography,[28,29] and 3D printing via two-photon polymerization,[30] face significant scalability and throughput limitations, restricting their practical adoption for large-area membrane production. Traditional UV-based photolithography methods offer better scalability, but their practical resolution limits (smallest pore sizes ~3 μm)[31] have prevented their use in precision immunoisolation membranes.[31,32]

Here, we develop a fabrication process to achieve rapid, wafer-scale laser writing of immunoisolation membranes with submicrometer pore sizes and high pore uniformity for precise

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immunoisolation. To achieve this unique combination of pore features, we use rapid laser scanning tools and engineer surface reflectivity and exposure parameters along with etching steps to shape pores with high geometric precision below the resolution limits of the exposure tools. At the submicrometer precision, we report membranes with substantially enhanced open pore area and size uniformity compared to existing membrane technologies. Finally, macrophage invasion studies demonstrate that these engineered membranes reliably prevent immune cell infiltration at pore sizes larger than those previously achievable. By enabling the rapid fabrication of membranes with improved pore density and pore size while maintaining stringent immune protection, the presented techniques represent a significant advancement in both microfabrication and immunoisolation technology.

RESULTS AND DISCUSSION

Using direct-write laser lithography, we developed membranes with uniform, submicron pores that are free of overlapping pore defects seen in previously used PCTE membranes, enabling more reliable immune cell exclusion at larger nominal pore sizes (Figure 1a). These membranes were fabricated using SU8 photoresists due to their chemical stability,[33] biocompatibility (among standard i-line resists already used in implantable devices),[33−35] and mechanical properties comparable with polycarbonates,[36−38] using a Heidelberg Instruments MLA 150 high-speed maskless aligner (a widely used system in both academic and industrial clean room facilities for rapid prototyping and microfabrication). Fast writing speeds are critical to enable a scalable and feasible microfabrication process compatible with the centimeter-scale sizes of most implantable systems. Our exposure times using MLA 150 in high-resolution mode were within ∼3−4 min/cm^2 (4−5 h for a full 4″, which is adaptable to larger wafer sizes), whereas our simulations, e.g., using an e-beam lithography tool (EBL Elionix HS50) resulted in ∼45−60 min/cm^2 to cross-link SU8 layers. This figure reached 20−60 min/cm^2 for a higher resolution direct laser-writing system (PicoMaster 200, Reith), depending on the selected resolution level (0.9−0.3 μm). Thus, direct-write laser lithography is significantly faster and more scalable than high-resolution alternatives such as e-beam lithography or two-photon polymerization and is compatible with batch wafer-scale processing. This combination of precision, scalability, and reproducibility offers a cost-effective pathway for translational applications.

While MLA 150 instruments offer high write speeds (10^2−10^3 mm^2/min), they struggle to achieve reliable submicron features. Furthermore, limitations on aspect ratio (<∼3−4) and focal depths (<∼3 μm) restrict the thickness of nanoporous membranes to only a few microns, resulting in fragile membranes that are susceptible to warping and fracture. We address these limitations via the fabrication workflow outlined in Figure 1b, which employs several engineering strategies for overcoming both resolution and mechanical challenges. To overcome resolution issues, we optimized the substrate reflectivity using a silicon nitride coating, CAD model tolerances, UV dose, and focal plane offsets. To improve mechanical integrity, we utilized a two-layer design where a thin membrane layer imparts immune exclusion capabilities, while a thicker but highly porous (80% open) backing layer provides mechanical stability. Additional critical processing steps include RIE etching to ensure open pores and a sacrificial layer that enables reliable lift-off. Subsequent sections detail the optimization of parameters for each step of the fabrication process and discuss the key failure modes. Careful implementation of these techniques is necessary to enable centimeter-scale free-standing nanoporous membranes (Figure 1c) with tunable mechanics and minimal effect on diffusion.

Sacrificial Layers for Lift-Off. In the absence of a sacrificial layer, the membranes fabricated through this process remained attached to the silicon wafers after SU8 was fully developed. Detaching the membranes from the silicon wafer by soaking in the resist removal solvents (i.e., *N*-methyl-2-pyrrolidone, NMP for 1 h) resulted in membrane permanent rolling shortly after lifting off (Figure S1a) due to the nonuniform swelling of the membrane layers. Hence, we considered using a sacrificial layer to lift the membranes after fabrication.

We note that although metal sacrificial layer candidates (coated via evaporation) provide high uniformity, the high reflectivity of metallic surfaces limits the minimum formable pore sizes (to ∼3 μm according to previous studies).[31] To achieve smaller pore sizes, we sought water-soluble sacrificial polymers, which are not soluble in organic solvents (to prevent the membrane layer from removing when spin coating the second SU8 layer and membrane rolling during lift off, Figure S1b). According to Linder et al.,[39] poly(acrylic acid) (PAA) offers high water solubility, organic solvent insolubility, and the most favorable coating uniformity among several other candidates such as poly(methacrylic acid), poly(vinyl alcohol), poly(ethylene oxide), *etc.* Another advantage of PAA as a sacrificial layer is its spontaneous covalent adhesion with SU8 (triggered by the acid groups in PAA). The adhesion of SU8 to the sacrificial layer is necessary for membrane layer attachment and stability during the spinning of the second SU8 support layer in our fabrication process.

Nonuniformities in Sacrificial Layer. Nonuniformities in the thickness of the sacrificial layer during the spin coating process lead to variations of surface reflectivity due to the thin-film interference effects (Figure S2a). These effect gradients are major hurdles in the wafer-scale photolithographic patterning of small feature sizes.[40] The reflection of the UV light at the PAA/SU8 interface back into the SU8 resist interferes with the light that travels through the SU8 resist, affecting the absorbed exposure dose and local pore size. Thus, any variation in the PAA thickness leads to pore size gradients across the wafer. As shown in Figure S2b, we computationally modeled reflectance from the acrylic thin films on silicon wafers as a function of PAA thickness based on Fresnel equations for the incident UV light λ = 375 nm. The results suggest periodic variations of reflectance between 0.27 and 0.57 (>∼100%) with PAA thickness changes as small as 60 nm.

As indicated in Figure S2c, an unbalanced spin coater can lead the PAA thickness to decline more than 30% from the center to the edges of the wafer. This thickness gradient results in radially periodic circular fringes (dark phase) wherein the pores were closed (Figure S2d,e) due to gradient interference effects. These periodic trends are in accordance with the reflectivity patterns computed in Figure S2b. At a small scale, we could observe these patterns around defects caused by dust-like particles on the wafer, leading to local defects in PAA and thereby fringe patterns (Figure S2f), highlighting the importance of clean microfabrication in this process.

As discussed, coating reproducibility and uniformity play a crucial role in repeatable patterning of SU8 across the wafer. PAA coating uniformity was enhanced (thickness gradient

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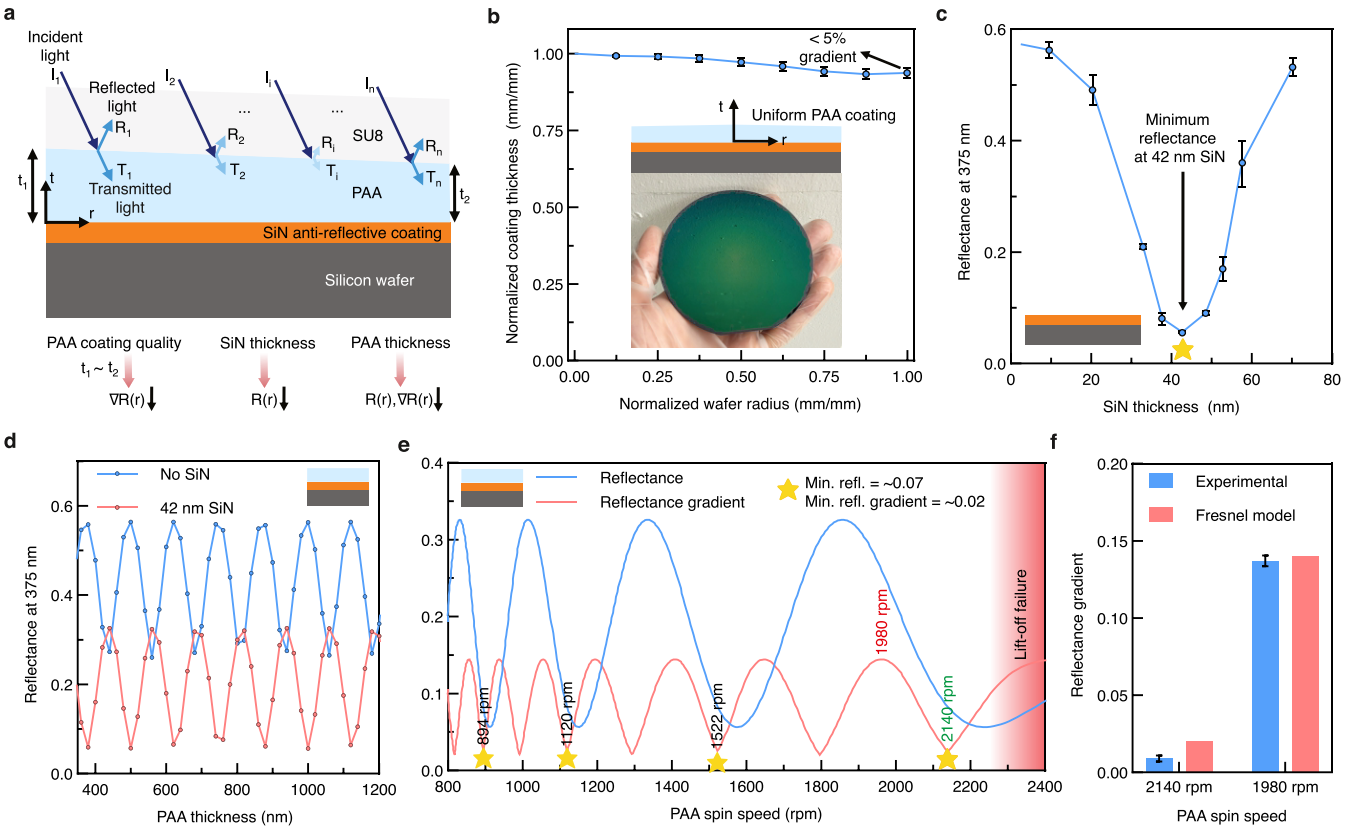


Figure 2. Engineering of surface reflectivity for optimized wafer-scale uniformity of microfabrication process. (a) Thin-film interference and effects of PAA uniformity on surface reflection. (b) Thickness profiles of PAA coatings obtained from a balanced spin coater. (c) Reflectance of silicon nitride (SiN) layers on silicon wafers at 375 nm. (d) Sensitivity of surface reflectance to PAA thickness variations with and without SiN antireflective coating using Fresnel modeling of thin-film interference. (e) Reflectance and its gradient as a function of PAA spin coating speed for minimizing surface reflection effects and thus exposure variations over a 4″ silicon wafer based on Fresnel modeling. (f) Experimental validation of reflectance gradients calculated via Fresnel analysis.

<5%) by spin coater lid adjustments and air pocket removal (vacuuming the spinner chamber), as shown in the thickness profiles of Figure 2a,b. Spin curves of Figure S3a suggest PAA film thickness control using spinning speed with standard deviations smaller than 15 nm. Thickness gradients at this optimized uniformity state ranged between 15 and 30 nm, regardless of spinning speed (Figure S3b), while this figure was greater than ~200 nm in the nonuniform state. This improved uniformity eliminated any visible fringe patterns and led to a uniform appearance in membranes, as shown in Figure S3c.

Despite these improvements, given the high sensitivity of substrate reflectance to the small PAA thickness variabilities (at the wafer-to-wafer and across-a-wafer levels), further strategies are taken as described below to optimize membrane reproducibility by minimizing the magnitude of reflectance (R) and its gradient (∇R) across a 4 in. wafer (Figure 2a).

Antireflective Strategies for Wafer-Scale Microfabrication of Small Pores. We considered using a bottom antireflective layer to minimize UV light reflectance and its gradients across the silicon wafers. With this approach, we aim not only to enhance pore size consistency but also to enable the formation of smaller pore sizes. For this purpose, a layer of silicon nitride (SiN) antireflective coating was deposited using plasma-enhanced chemical vapor deposition (PECVD). Figure 2c demonstrates the effects of SiN thickness on reflectance at 375 nm wavelength measured experimentally by using a spectrophotometer. Here, the minimized reflectance was achieved at 42 nm film thickness, which was formed after an 80 s deposition time (Figure S4a).

The trends of Fresnel substrate reflectance for a double layer of PAA coating on top of a SiN layer are shown as a function of the PAA thickness in Figure 2d. According to the results, introducing the SiN layer significantly reduces the range of reflectance from 0.27−0.57 (Figure S2b) to 0.05−0.32. Despite the magnitude of reflectance being reduced with SiN remarkably, the PAA thickness gradients can yet result in up to ~14% reflectance variations from the wafer center to the edges (Figure S4b) for a maximum of 25 μm *inevitable* PAA thickness variations (according to Figure S3b) due to the spin coating process. Although this upper limit is still smaller than those without SiN antireflective layers, we further optimized the PAA spin parameters to minimize reflectance gradients due to these inevitable PAA thickness decays. For this purpose. The reflectance (R)-PAA thickness (t_{PAA}) relationship shown in Figure 2d is estimated by eq 1

$$R(t_{PAA}) = A\cos(Bt_{PAA} - C) + D \quad (1)$$

Given that PAA thickness decreased by an approximately constant amount (Δt = 25 nm according to Figure S3b) continuously from the wafer center ($t_{PAA,c}$) to edges ($t_{PAA,c} - \nabla t$), we calculated the reflectance gradient corresponding to PAA thicknesses at the wafer center, $\nabla R(t_{PAA,c})$, based on the relative difference between maximum and minimum reflectivity over the thickness gradient range as seen in eq 2

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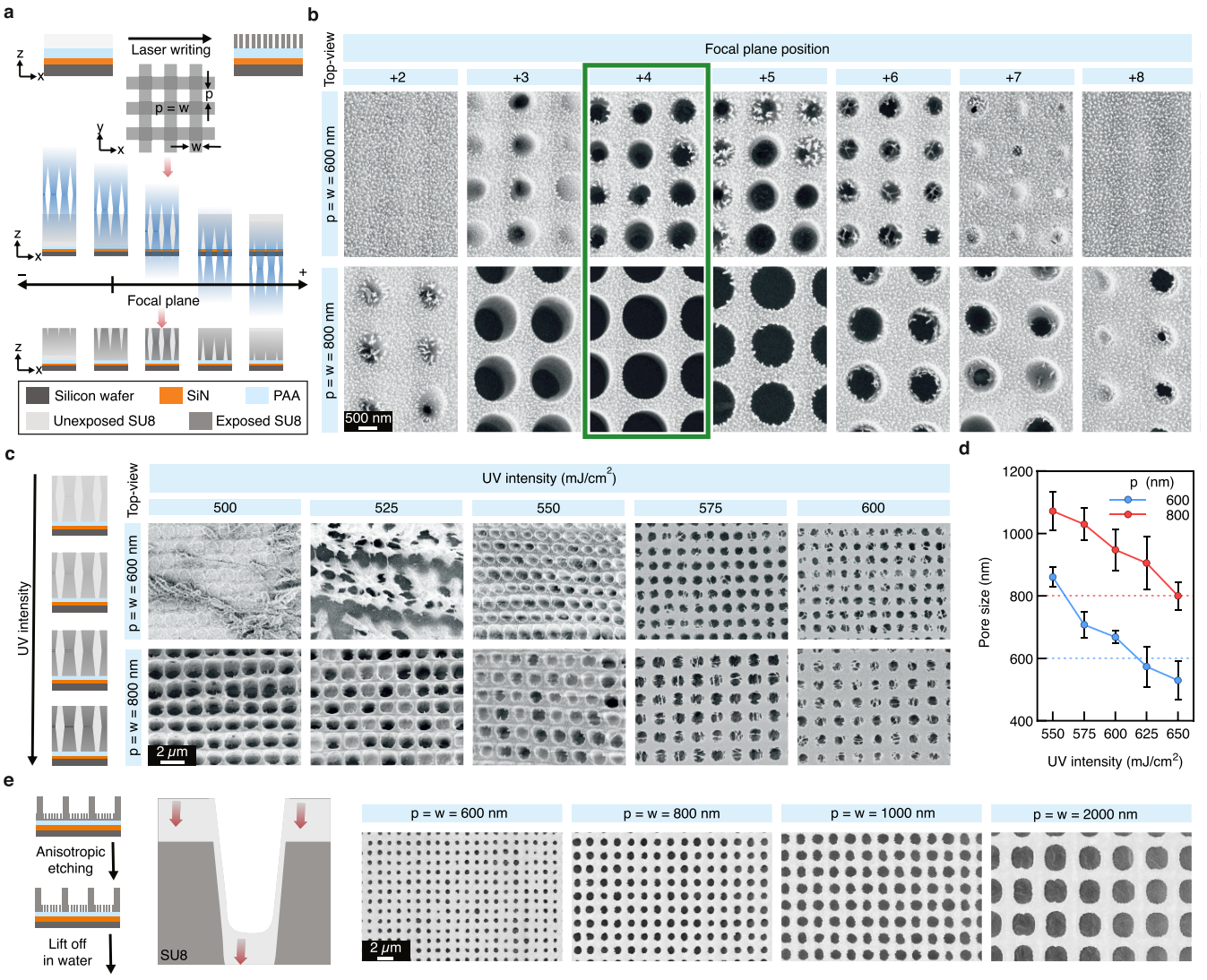


Figure 3. Optimization of the microfabrication process for control of pore characteristics. (a) The input computer-aided design (CAD) model determines pore size (p) and wall thickness (w). The laser focal plane is tuned for writing open pores to avoid closed pores due to out-of-focus exposure. (b) Top-view scanning electron microscopy (SEM) image of membranes exposed at various focal plane settings (after reactive ion etching, RIE). (c) Effects of exposure dose on pore morphology. The SEM images represent a top-view of membranes before RIE etching. (d) Pore size control via UV intensity. (e) Anisotropic etching (Bias RIE) for complete opening of pores with sizes as small as 600 nm (designed size) in membranes fabricated on reflection-controlled wafers (using silicon nitride, SiN, coatings). The SEM images show the bottom view of the membranes.

$$\nabla R(t_{\mathrm{PAA,c}}) = \frac{\max(R)([t_{\mathrm{PAA,c}} - \Delta t,\ t_{\mathrm{PAA,c}}]) - \min(R)([t_{\mathrm{PAA,c}} - \Delta t,\ t_{\mathrm{PAA,c}}])}{\min(R)([t_{\mathrm{PAA,c}} - \Delta t,\ t_{\mathrm{PAA,c}}])} \quad (2)$$

The reflectance gradient on a 4″ silicon wafer as a function of PAA thickness showed periodic local minimums (Figure S4b). Based on the experimental spin curves of PAA (Figure S3a), this reflectance gradient was plotted as a function of spin curves, where those local minimums correspond alternatively to high and low reflectance, as shown in Figure 2e. Here, we seek spin speeds where ∇R is minimized, among which those resulting in the lowest R (e.g., 2140 rpm, 1522 rpm, 1120 rpm, 894 rpm, *etc.*) are of interest. This result shows that rational selection of PAA thickness at the center using spin speed control minimizes reflectance gradients down to 2% and reflectance to ∼7%. As shown in Figure 2f, the experimental reflectance profile over the wafer radius for a PAA coating at a spin speed of 2140 rpm (i.e., at $t_{\mathrm{PAA,c}}$ = 393 nm) resulted in 2% reflectance gradient across the wafer in agreement with Fresnel analysis, while this figure reached ∼14% when spin speed was 1980 rpm.

Due to better thickness control, higher spin speeds among the optimum set of spin speeds were more favorable. However, we note that higher spin speeds that led to smaller PAA thicknesses than 393 nm (speeds of >2140 rpm) failed to lift off the membranes in water, as the reactive ion etching (RIE) step led to the permanent attachment of membranes to silicon wafers. Thus, a spinning speed of 2140 rpm was used to minimize reflectance and its gradients for uniform exposure across the wafer.

Optimization of Laser-Writing Parameters for Membrane Patterning. The laser exposure parameters, including dose and focus, were optimized using the membrane

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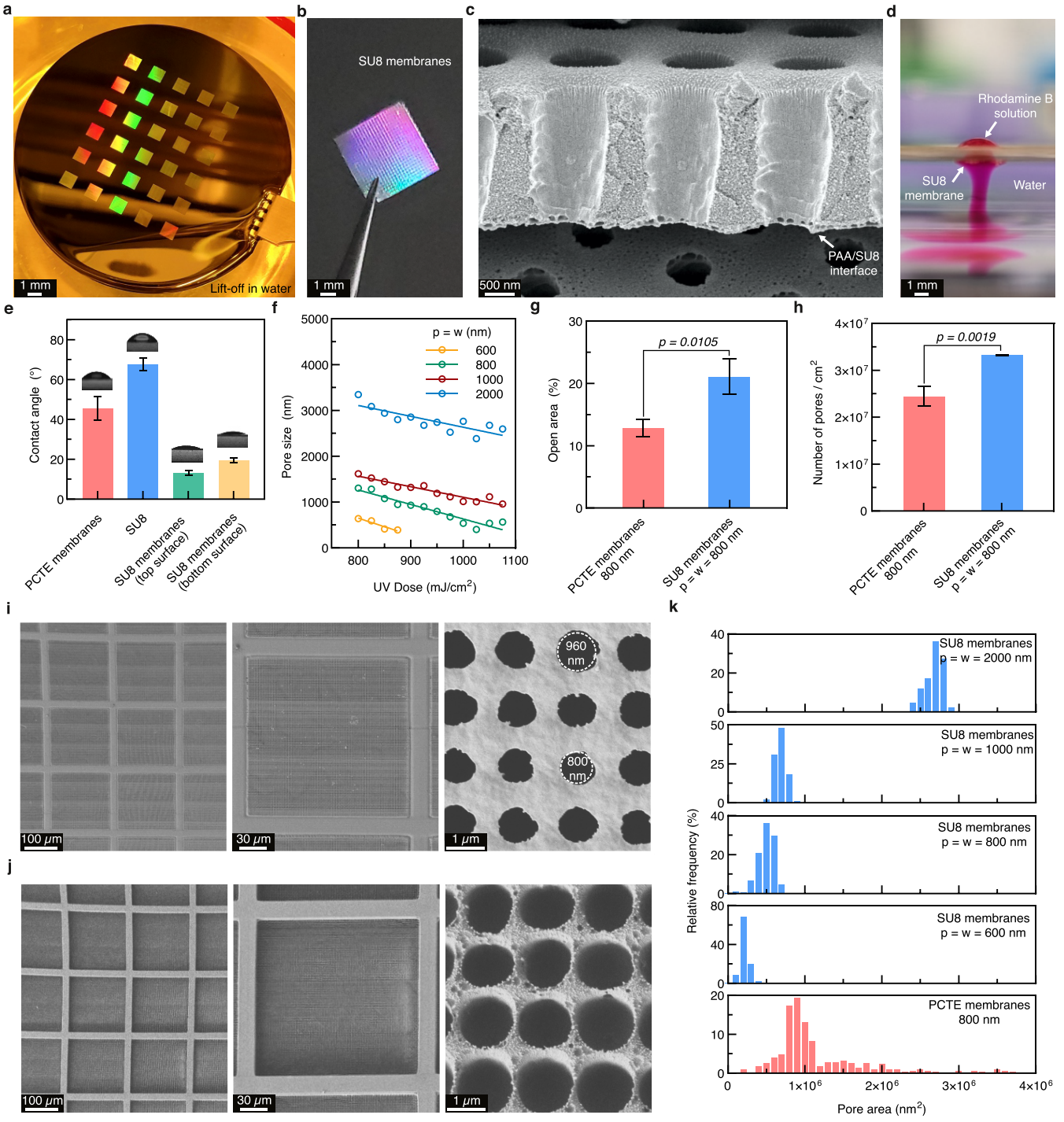


Figure 4. Characterization of microfabricated SU8 membranes. (a) Membrane lift-off in water. (b) Image of dried SU8 membrane after lift off. (c) Cross-sectional scanning electron microscopy (SEM) image of the SU8 membrane designed with pore size (p) and wall thickness (w) of 800 nm after the bias reactive ion etching (RIE) process. (d) Rhodamine B solution passing through a sandwiched membrane between punctured plastic tapes floating on the water surface, suggesting open pores capable of small molecule transfusion. (e) Comparison of contact angle results for commercial polycarbonate track-etched (PCTE) membranes with SU8 and SU8 membranes. (f) Controllability of pore size with UV exposure parameters for membranes after RIE treatment. (g) Open area and (h) pore density in SU8 membranes (p = w = 800 nm) and PCTE membranes with 800 nm pore size. Low- and high-magnification SEM images of SU8 membranes ($p = w$ = 800 nm) from (i) bottom, and (j) top views. (k) Pore area histograms obtained based on the bottom-view SEM images for membranes with various size designs.

computer-aided design (CAD) model. The membrane CAD designs consisted of a perpendicular array of gratings (instead of patterned pores), allowing for less CAD file size and faster toolpath generation (Figure 3a). The pore size (p) and pore wall thickness (w) were equal ($p = w$), yielding a nominal open area of 25%. As a first step, the focus plane of exposure was identified because misfocus could cause pore closure, as shown in Figures 3b and S5. Direct writing at various focal planes has

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been performed, as shown in Figure 3b, where the focus setting of +4 resulted in the best pore formation. Then, the effect of the exposure dose on the pore diameter before RIE was studied (see Figure 3c). Cross-linking of SU8 was obtained at UV doses larger than 575 mJ/cm^2.

We note that following the standard SU8 postexposure baking protocols (i.e., 1 min at 80 °C and 2 min at 110 °C)[41] resulted in pore closure in membranes with submicron pores due to SU8 over polymerization (Figure S6). It was necessary to reduce the baking temperature to 65 °C (4 min) for the open pore formation. The spinning of the secondary support SU8 layer did not cause any damage to the membrane layer due to its robust adhesion to the PAA sacrificial layer. The pre-exposure baking of this secondary support layer was performed at lower temperatures (65 °C) than those specified in standard protocols[41] but for a longer duration (15 min) to avoid pore closure in the membrane layer.

In addition to the above considerations, we found that SU8 development and rinsing steps after the final baking step play critical roles in uniform and open pore formation across the wafer (Figure S7). Particularly for lower (submicron) pore sizes, while the absence of a rinsing process with isopropanol (IPA) led to fully closed pores, direct immersion of silicon wafers from SU8 developer to IPA resulted in random pore closures on the wafer. We found that a gradual solvent exchange process from SU8 developer, propylene glycol monomethyl ether acetate (PGMEA), to IPA was necessary to ensure uniformly open pores throughout the wafer.

Reactive Ion Etching for Consistent through Pores. The RIE etching before lift-off in water was crucial in obtaining open pores and preventing membrane cracking while drying after lift-off. Figure S8 compares the results of membranes fabricated with and without RIE etching. Without RIE etching, as shown in Figure S8a, water surface tension while drying after lift-off resulted in pulling the membranes toward the open windows of the support grid, causing the thin membrane layers to rupture. This rupture was prevented by oxygen plasma treatment during the RIE etching process (Figure S8b) due to enhanced hydrophilicity, mitigating high surface tension effects.

In addition, after inspecting the membranes without RIE treatment, we noted a cross-linked continuous thin layer of cross-linked SU8 at the interface with PAA, which led to closed pores in membranes (Figure S8c). This pore closure is attributed to the spontaneous cationic polymerization of SU8, triggered via acid groups in PAA and RIE etching was able to unclog these pores, as shown in Figure S8d. We note that as a potential solution, we also tested neutralizing PAA solutions before spin coating. Although this approach resulted in open pores, the membranes were found to form wrinkling defects and fractures (Figure S9) due to the lack of adhesion when spinning the secondary SU8 support layer.

Given the need for plasma etching steps for open pore formation and crack prevention, we tested different approaches of isotropic etching using a plasma asher and anisotropic etching using a dry RIE process for different exposure times on wafers without SiN coatings (Figure S10). According to Figure S10a,b, all pores were closed without plasma treatment. Figure S10c,d shows the top and bottom views of the membranes after isotropic etching process, respectively. Isotropic etching of more than 1 min resulted in significant damage to pore walls, particularly as the pore size (and pore wall thickness) decreased. Among the different design pore sizes, only those with $p = w = 2000$ nm resulted in open pores via isotropic plasma etching. In our subsequent experiments, we evaluated anisotropic plasma etching using a biased RIE as illustrated in Figure S10e,f. We found no significant damage to the membrane pore walls in all pore size groups over a 5 min etching time. The bottom-view images demonstrated the formation of open pores in designs with $p = w$ at values as low as 800 nm through anisotropic etching.

To highlight the role of antireflection coatings, we performed the same anisotropic RIE etching on membranes fabricated on silicon wafers with SiN (see Figure 3e) to compare with those without a SiN layer (i.e., Figure S10e,f). The results suggest that using SiN layers enables the formation of much smaller pore sizes as low as p = w = 600 nm.

We remind that according to the cross-sectional images (Figures S5 and S8c) pores can form a dome-shaped structure before RIE etching due to unfocused exposure or unwanted SU8 cross-linking possibly because of PAA-triggered polymerization or surface reflectivity. The RIE etching of dome-shaped pores results in wedge-like pore wall morphologies (Figure S8d), which aids in the formation of smaller pore sizes (<600 nm) at the opposite side of the membranes, allowing for pore sizes even below the exposure tool resolution (i.e., ∼1 μm).

Characterization of Membranes. Here, the final membranes fabricated by the optimized process are characterized. After the RIE process, membranes are lifted off in water (Figure 4a) and dried (Figure 4b). A high-magnification image of the membrane cross-section shows open ∼800 nm pores with aspect ratios of ∼2, and a thin porous layer at the bottom of the membrane where it comes in contact with the PAA (Figure 4c). Open pores were further demonstrated by passing a dye solution (Rhodamine B) through the membrane, as illustrated in Figure 4d. The porous layer at the interface with PAA suggests chemical incorporation of PAA onto the SU8 surface. Covalent presence of PAA on this side and oxygen plasma treatment of the opposite side of the membrane (due to RIE etching) enhances membrane hydrophilicity compared to SU8 controls, surpassing those of commercial PCTE membranes, according to water contact angle results of Figure 4e. Apart from facilitating crack-free lift off, the improved hydrophilicity also allows for efficient transfusion of small molecule therapeutics through the pores.

Control on the membrane pore size is not only achieved via the input CAD model, but also the exposure dose could be optimized for further tuning of pore size (Figure 4f). It is noted that the UV dose required for matching the pore size with that designed in the CAD model increases as the CAD pore size (=pore wall thickness) increases. The open area and pore density of the membranes with 800 nm designed pore size is compared to the commercially available membranes in Figure 4g,h, respectively. The SU8 membranes at the same pore sizes as polycarbonate track-etched (PCTE) membranes posed a significantly larger (∼40%) open area due to more packed pores (∼70%) enabled by the elimination of unetched regions and overlapped pores. The open area of ∼22% in SU8 membranes is also much larger than that in commercially available microfabricated SiN membranes (SimPore, Inc.) with 15% open area.

The low magnification SEM images of membranes (design parameters: $p = w = 800$ nm) from both sides are presented in Figure 4i,j. These images demonstrate crack-free and ordered distribution of pores in millimeter-scale windows, where smaller pores on the bottom view (interface with silicon

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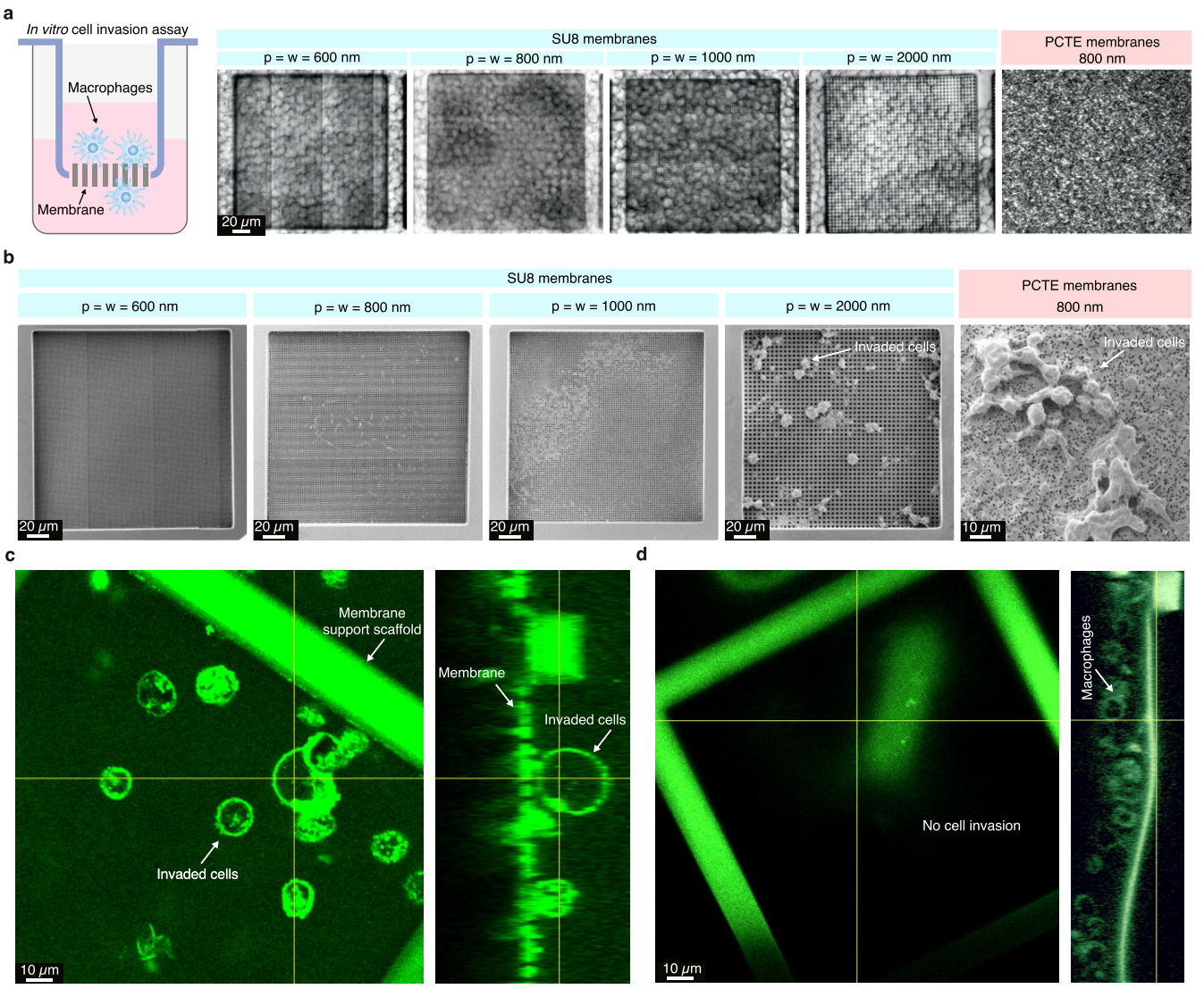


Figure 5. ***In vitro*** **cell isolation performance of the microfabricated SU8 membranes using macrophages to assess immunoisolation performance. (a) Bright-field images of macrophages cultured on polycarbonate track-etched (PCTE) and SU8 membranes designed with various pore sizes (p) and wall thicknesses (w). (b) Scanning electron microscopy (SEM) images of invaded macrophages 7 days postculture. (c, d) Confocal images of invaded cells for SU8 membranes with p = w = 2000 nm and p = w = 1000 nm, respectively.**

wafer) compared to top-view are reflective of wedge-like pore walls as discussed above. The periodic vertical stripes visible in the pictures are aligned with the head-write exposure toolpath and are due to nonuniformities of UV light across the laser exposure field, which led to the pore size distributions around the designed pore size, as shown in Figures 4k and S11a. The pore size peaks matched well with the designed pore sizes, and size distributions were more symmetric at larger design pore sizes (p = w = 1000 and 2000 nm) as compared with 800 and 600 nm. In comparison with commercial PCTE membranes (800 nm), SU8 membranes demonstrated narrower pore size variations as a result of more ordered pore placement, eliminating pore overlapping, suggesting a more precise pore control in polymeric membranes. For a better understanding of pore morphology, the circularity of pores in membranes is analyzed in Figure S11b. It is noted that as the design pore size decreases toward the resolution limit, the pore eccentricity increases, likely due to the differential errors of laser-writing speeds in x and y directions at small scales. Furthermore, overlapping pores in PCTE membranes resulted in a scattered distribution, highlighting a more uniform shape distribution in SU8 membranes.

Cell Isolation Performance. Immunoisolation is a critical requirement in the implantable devices functioning based on active biological components such as cell macro-encapsulation for cell therapies. Hence, the biofiltration capability of SU8 membranes was studied in terms of their immunoisolation performance by culturing macrophages on the flat side of the membranes (Figure 5). To better mimic physiologically relevant driving forces for immune cell migration, the transwell culture setup was configured to apply both a hydrostatic pressure differential and a chemotactic gradient. This was achieved by adding a larger volume of medium containing macrophage colony-stimulating factor (m-CSF) to the lower chamber, which encouraged macrophage movement toward the underside of the membranes during incubation.

Membranes were inspected and remained stable during incubation under physiological conditions. Cell evaluation via bright-field microscopy revealed that as opposed to the PCTE membranes, the SU8 membranes provided transparency for

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viewing and monitoring the cell population (Figure 5a). The results further showed high cell populations on all membranes 7 days postculture, supporting cell proliferation and viability. To investigate the immune cell invasion, SEM images were obtained from the opposite side of the membranes after 7 days (Figures 5b and S12). In the commercial PCTE groups, the results indicated cell invasions from 800 nm membranes, while no cell invasion was seen in SU8 membranes with larger pore sizes ($p = w = 1000$ nm). Cell invasions in PCTE are attributed to the overlapping pores due to their random pore distribution. We note that immunoisolation size thresholds seen here are comparable to those reported previously *in vivo*.[25,42,43] The ordered and open-through-pore architecture achieved here in membranes at the immunoisolation threshold size of 1 μm enabled both high filtration precision and maximized open area without risking immune cell penetration.

Interestingly, although macrophages are as large as 10 μm in diameter based on the optical images of Figure 5a, they were able to infiltrate through pores of an order of magnitude smaller than their own sizes. We further studied cell infiltration via live cell fluorescent imaging (see Figure 5c). The results show cross sections of infiltrated cells migrated through 2 μm pores with no infiltration through 1 μm SU8 membranes. While these images agree with the SEM results, they reveal the highly plastic nature of macrophages and their infiltration mechanism through the amoeboid migration process. During this process, cells dynamically reorganize their actin cytoskeleton, allowing them to transiently alter their shape and reduce their cross-sectional area to fit through constricted spaces. Such cytoskeletal plasticity enables the passage of the cell body, including the relatively large nucleus, through pores significantly smaller than the cell's apparent diameter.

Study Limitations. Proof-of-concept studies revealed the potential of microfabricated membranes for precise and robust immunoisolation. However, *in vivo* application of these membranes when integrated with implantable devices is yet to be studied. This integration is particularly critical in cell microencapsulation devices for glycemic control in patients with Type I diabetes. Furthermore, antifibrosis is a crucial aspect in implantable applications involving drug transfusion. The fabrication process presented here enables implementing chemistries that allow for surface immobilization via antifibrotic and immunomodulatory building blocks, which further strengthen their translational potential. Although this study focuses on pore control, further advancements in improving the mechanical resilience of these membranes are essential to ensure their long-term durability when implanted in dynamic tissues. While the pore size and distribution effects on immunoisolation are revealed, the presented methodology enables future studies on understanding the roles of pore shape and irregularities on the immune response and filtration of these membranes.

CONCLUSIONS

Membranes with precisely controlled submicrometer pores are essential for immunoisolation applications, particularly in implantable medical devices where selective molecular exchange is required without immune cell infiltration. Formation of nonoverlapping pores at sizes that ensure immunoisolation requires microfabrication techniques; however, current methods entail long exposure times, hindering their scalable fabrication. In this study, we demonstrated a rapid and scalable laser microfabrication approach to produce SU8 membranes with highly controlled submicron pores for precise immunoisolation. By leveraging a water-soluble sacrificial layer, we achieved reliable lift-off while maintaining mechanical integrity through a two-layer SU8 structure. The introduction of antireflective coatings and optimized exposure conditions enabled the formation of pores below the resolution limit of conventional laser aligners, eliminating the risk of oversized pores due to random overlaps observed in track-etched membranes.

Reliable exclusion of immune cells depends on every pore meeting the intended cutoff dimension, highlighting the need for precise control over pore size and placement. In contrast, conventional track-etched membranes exhibit random pore arrangements that can result in overlapping pores and oversized channels, which compromise the filtration reliability. The ordered pore architecture achieved here enables both high filtration precision and a maximized open area without risking immune cell penetration. Our results highlight the critical role of substrate reflectivity, exposure parameters, and plasma etching in achieving precise pore morphology at the immunoisolation level. The optimized process yielded membranes with a high pore density and increased open area compared to conventional PCTE membranes. *In vitro* macrophage studies demonstrated effective immunoisolation with well-defined pore size thresholds, preventing immune cell infiltration at larger pore sizes than conventional membranes, which combined with a high open pore area, holds promise for applications in immunoprotected cell therapies.

METHODS

Membrane Microfabrication Process. First, 42 nm silicon nitride layers were deposited on 4″ silicon wafers (UniversityWafer Inc., type P, ⟨100⟩ orientation) via plasma-enhanced chemical vapor deposition (PECVD) (PECVD System PD-220, Samco Inc.) for 80 s using RF power of 125 W under 7 sccm SiH_4, 6 sccm NH_3, and 197 sccm N_2 flow. Upper and lower heater temperatures were set to 150 and 350 °C, respectively. Next, the wafers were treated with oxygen plasma using an AutoGlow 1000 asher (Glow Research AutoGlow) at 200 W for 3 min to facilitate the coating of a water-soluble sacrificial layer. To deposit the sacrificial layer, poly(acrylic acid) (PAA), average Mw ∼100,000, 35 w/v% in water (Sigma-Aldrich, 523925) was diluted to 10 w/v% using ultrapure water (Fisher Scientific). A 10 mL aliquot of the diluted PAA solution was dispensed onto the silicon wafer and spin-coated (CEE Apogee) at 2000 rpm for 15 s, followed by baking on a hot plate at 110 °C for 4 min.

For the membrane layer, SU8 2005 (Kayaku Advanced Materials, Inc.) was spun at 3000 rpm for 30 s, following an initial spinning step at 500 rpm for 5 s. Pre-exposure baking was performed by placing the wafer on an 80 °C hot plate for 1 min and then transferring it to 110 °C for an additional 2 min. The membrane layer was exposed using an MLA 150 maskless laser aligner (Heidelberg Instruments) with a 375 nm UV laser using orthogonal grid patterns where the pore size matched the pore wall thickness. Designed pore sizes of 600, 800, 1000, and 2000 nm were exposed at optimized dose and defocus settings using pneumatic focus and a standard depth of focus. Postexposure baking was performed at 65 °C for 4 min.

A secondary mechanical support layer was formed by spinning a thick SU8 2010 (Kayaku Advanced Materials, Inc.) at 500 rpm for 5 s, followed by spinning at 2000 rpm for 30 s. The wafer was pre-exposure baked at 65 °C for 14 min, and a grid pattern consisting of 20 μm thick orthogonal lines with 200 μm spacing was exposed following the same procedure as the membrane layer, but with a UV intensity of 1000 mJ/cm^2. The final postexposure baking was conducted at 65 °C for 4 min.

To develop the SU8, the wafer was agitated in propylene glycol monomethyl ether acetate (PGMEA) for 15 min, with the PGMEA

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solution refreshed after the first 2 min. To transition from PGMEA to isopropyl alcohol (IPA), the wafer was first covered with fresh PGMEA. IPA was then gradually added while agitated for 10 min, followed by immersion in an IPA bath for another 10 min. After drying overnight, the membranes were treated via reactive ion etching (RIE) under a 50 sccm oxygen flux, 150 W bias power (no ICP power) for 240 s at 5 Pa chamber pressure. Finally, the wafers were soaked in DI water, leading to the spontaneous lift-off of the membranes, and dried in air.

Scanning Electron Microscopy. The samples were affixed to a glass slide. Then, a 20 nm gold layer was sputtered by using a Blazers SCD 040 sputtering system. Scanning Electron Microscopy (SEM) images were taken using a Zeiss Gemini 450 electron microscope at 1 kV accelerating voltage.

Surface Reflectance and Film Thickness Measurements. Surface reflectance and coating measurements were performed by using a Filmetrics F20 thin-film analyzer. Reflectance data was collected in the wavelength range 350−850 nm.

Contact Angle Tests. A Krüss drop shape analyzer was used to take and analyze the contact angle images of 10 μL drops of deionized water on the surface of the membranes.

***In Vitro* Cell Invasion Tests.** Membranes were sandwiched between Scotch tapes, which were punched using a 3 mm biopsy punch. The original membranes of the transwell inserts in Corning Transwell 24-well plates (Sigma-Aldrich, CLS3413−48EA) were removed and replaced with the sandwiched microfabricated membranes via the adhesive side of the Scotch tapes. For sterilization, 1 mL of 70% (v/v) ethanol was added to the wells containing the modified inserts and incubated for 30 min, followed by phosphate-buffered saline (PBS) washing and air drying to prepare for cell culture experiments.

Tumor-associated macrophage cell lines (RAW264.7) were cultured in the transwells at a density of 9×10^4 cells per well. The cells were plated on the flat side of the membranes (because the membranes were attached with the support layer facing down) and incubated in a Forma Series II incubator (Thermo Fisher) at 37 °C with 5% CO_2. The culture medium, Dulbecco's Modified Eagle Medium (DMEM), was refreshed every 2 days. To promote active cell infiltration under physiologically relevant conditions, a hydrostatic pressure differential and chemotactic gradient were introduced in the transwell setup. Specifically, 300 μL of culture medium was added to the upper chamber (cell side) and 500 μL to the lower chamber, creating a downward pressure gradient favoring cell movement toward the bottom compartment. The medium in the lower chamber was supplemented with 20 ng/mL macrophage colony-stimulating factor (m-CSF) to establish a cytokine gradient that further stimulated macrophage migration across the membranes. Cell invasion was assessed via fluorescence imaging using CellBrite Steady 488 cell membrane staining dye (Biotium, #30106). The staining solution was prepared by diluting the dye in medium (2 μL of dye per 1 mL of medium), and the cells were incubated in this solution for at least 30 min before imaging. Imaging was performed by placing the transwells on 12-well plates with No. 1.5 coverslips (MatTek Life Sciences, P12G-1.5−10-F) and using an Olympus FV1200 laser scanning confocal microscope at 505/529 nm excitation/emission wavelengths with 0.5 μm z-stack increments. Images were analyzed by using ImageJ 1.54f software.

Cell invasion was further analyzed by SEM. Cells were fixed in 10% (v/v) formalin for 30 min and washed with PBS. The membranes were then frozen in liquid nitrogen and freeze-dried for 2 days. The membranes were imaged from the side with the visible support structure to inspect for invaded cells.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.5c09814.

Optimization of membrane lift-off, characterizations of the coating nonuniformity during the spin coating, thickness profiles of PAA sacrificial layers, surface reflectivity of SiN layer, cross-sectional image of membranes when exposure is out-of-focus, effects of baking temperature, optimization of solvent exchange process, and RIE on formation of open uniform pores, effects of PAA's pH, characterizations of pore size distribution and pore noncircularity in membranes, and quantification of cell invasion studies (PDF)

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Author Contributions

The original idea was developed, and experiments were designed by H.M. H.M. performed the majority of experiments, conducted data analysis, and prepared the original draft of the manuscript. W.J.J. assisted with *in vitro* cell invasion studies and contributed to fabrication process optimization. Y.Z. performed experiments to optimize oxygen plasma etching processes. R.S.M. assisted H.M. in contact angle measurements. D.G.A., S.R.K., and M.A.B. contributed to idea development and experimental design. D.G.A. and R.S.L. supervised, conceived of this study, and all authors contributed to the writing of the manuscript. The final version of the manuscript has been approved by all authors.

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Notes

The authors declare the following competing financial interest(s): D.G.A. discloses potential competing interests due to his past and present advising in, Sanofi, Alnylam, Translate Bio, oRNA Therapeutics and Combined Therapeutics. R.L. discloses potential competing interests due to his advising for a list of entities provided in Supporting Information.

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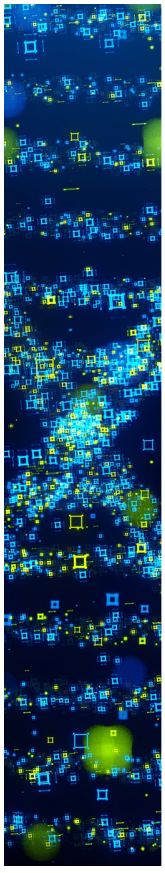
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